Trend × Volatility Market Regime Detection

Complete Model Manual

Concepts, Mathematics, Implementation and Interpretation

Asset	SPY – S&P 500 ETF (Yahoo Finance)
Calibration	2000 to 2020 (in-sample)
Evaluation	2021 to present (out-of-sample)
History	6,383 trading days from Oct 2000 to Mar 2026
Regimes	6 distinct states
Implementation	Python 3.10+, pandas, numpy, matplotlib
Current Signal	**RISK_ON_FRAGILE**, Confidence 88.8% (2026-03-05)

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1 Purpose and Philosophy

1.1 What Is a Market Regime?

A *market regime* is a persistent state in which financial markets behave in a structurally similar way. During different regimes, the same asset exhibits very different statistical properties: return distributions shift, volatility clusters, correlations between assets change, and the risk/reward balance that investors face changes fundamentally.

Think of it like weather. Knowing that today is Tuesday tells you almost nothing about whether to carry an umbrella. But knowing that you are currently inside a storm system (a *persistent state*) tells you a great deal. Market regimes work the same way. Individual days are noisy. Persistent structural states are informative.

Regime vs. Day

A regime is not a prediction about tomorrow. It is a description of the environment the market is currently operating in. You are not asking "Will the market go up?" You are asking "What kind of market is this?"

1.2 Why Not Just Predict Returns?

The natural instinct in quantitative finance is to build a model that forecasts tomorrow's return. This approach faces a fundamental problem: **daily stock returns are nearly unpredictable**.

Decades of academic research confirm that the signal-to-noise ratio in daily equity returns is extremely low. The autocorrelation of S&P 500 daily returns is statistically indistinguishable from zero. Even sophisticated machine learning models, trained on vast amounts of data, struggle to produce consistent out-of-sample alpha from short-horizon return prediction.

Return prediction requires forecasting the precise direction and magnitude of price movement. Regime detection sets a much more achievable goal: identifying the *environment* the market is operating in. This is a coarser question, but a far more answerable one.

1.3 What This Framework Provides

Method 1 is designed as a *base layer*, which is a structural reading of market conditions that can inform position sizing, risk controls, and higher-level model inputs. Its defining properties are:

- **Deterministic:** Every regime assignment traces directly to the price and volatility values that day. No randomness, no model estimation uncertainty.
- **Look-ahead free:** No future data enters the computation at any point, including in the confidence score.
- **Interpretable:** Any practitioner can open the code and understand exactly why a given day received a given label.
- **Extensible:** The regime label and confidence score are clean scalar outputs on every trading day. They serve as direct inputs for portfolio overlays or more complex models.

2 The Two Dimensions: Trend and Volatility

2.1 Why These Two Variables?

Of all the variables one could use to describe market conditions, **trend** and **volatility** are chosen because together they capture the two most economically meaningful dimensions of equity market behaviour.

Both are directly observable from price alone. No analyst estimates, no earnings data, no macroeconomic forecasts. This makes the framework applicable across asset classes and time periods without requiring data that may not be available historically.

2.2 What Each Dimension Measures

Trend measures the direction of long-term capital flows. When large institutional investors (pension funds, sovereign wealth funds, asset managers) are in aggregate allocating capital toward equities, prices trend upward. When they are de-risking, prices trend downward. Trend is the structural, slow-moving component of price behaviour.

Volatility measures the level of uncertainty, stress, and disorder in those capital flows. High volatility does not simply mean "the market is moving a lot". It means investors are uncertain, risk limits are being breached, leveraged positions are being forced to close, and correlations between assets are rising. Volatility is the stress component.

2.3 Orthogonality: Why They Are Independent

Orthogonal means statistically independent, meaning knowing one tells you nothing about the other. This is not a theoretical assumption; it is an empirical fact observable in market data. Table 1 shows four real historical combinations.

Table 1: The four distinct environments produced by combining trend and volatility

Trend	Volatility	Real-world example
UP	LOW	2017: steady, quiet advance, the calmest year in SPY history
UP	HIGH	2020 recovery: violent upside after the COVID crash
DOWN	LOW	2015–16: slow, quiet de-risking as China concerns mounted
DOWN	HIGH	2008 GFC: panicked selling with extreme daily swings

A model built on only one dimension would collapse these four states into two, and that means losing critical structural information. A calm uptrend and a violent uptrend require completely different portfolio behaviours even though both are "going up."

3 Data and Input

3.1 The Price Series

The only input to this model is a **daily closing price series**. All features are derived from this single time series. Closing prices are used because they represent the final, settled price at which the market agreed to transact on a given day. It is the most widely used and highest-quality daily price signal.

Data is downloaded via the `yfinance` library (Yahoo Finance). The minimum required history is 2,000 trading days (approximately 8 years), because:

- The 200-day moving average requires a 200-day burn-in before producing its first valid value.
- The calibration window (2000–2020) must contain sufficient data for stable quantile estimates.

3.2 Why SPY?

SPY (SPDR S&P 500 ETF) is the primary instrument because it is the most liquid equity instrument in the world, with over 25 years of clean daily price history, no survivorship bias, and behaviour that is representative of the broad US equity market.

Applying to Other Assets

The framework is not SPY-specific. It can be applied to any instrument with sufficient history. However, the volatility thresholds are calibrated on SPY's distribution and **must be recalibrated** when applying to other assets. A small-cap ETF with 25% long-run volatility has a fundamentally different distribution than SPY at 15%.

4 Feature Engineering

Feature engineering is the process of transforming raw data into inputs that capture the structural properties the model needs. Four features are derived from the price series. The implementation lives in `features.py`.

4.1 Log Returns

The daily log return on day t is defined as:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right) \tag{1}$$

where P_t is the closing price on day t.

Why log returns instead of simple returns?

Simple returns are defined as $R_t = (P_t - P_{t-1})/P_{t-1}$. Log returns are preferred for three reasons:

1. **Additivity over time.** The multi-period log return is the *sum* of individual log returns:

$$r_{t_1 \to t_n} = \sum_{i=t_1}^{t_n} r_i$$

   Simple returns are multiplicative and harder to aggregate.

2. **Symmetry.** A 50% loss followed by a 100% gain returns to the starting point in log-return space: $\ln(0.5) + \ln(2) = 0$. Simple returns do not have this property: $-50\% + 100\% \neq 0$.

3. **Approximate equivalence for small changes.** For the small daily moves typical in equity markets, $\ln(1 + R_t) \approx R_t$, so both measures agree in practice.

Important: In this framework, log returns are used as inputs to the volatility computation, not as directional signals. The return series itself is never used to predict direction.

Listing 1: Log return computation in features.py

```
df["ret"] = np.log(df["price"] / df["price"].shift(1))
```

4.2 Realised Volatility (20-Day)

Realised volatility is an estimate of the true underlying volatility of a price series, computed from observed returns. The 20-day annualised realised volatility is:

$$\sigma_{20}(t) = \sqrt{252} \cdot \text{std}(r_{t-19}, r_{t-18}, \ldots, r_t) \tag{2}$$

where $\text{std}(\cdot)$ is the sample standard deviation of the 20 most recent log returns.

Why 20 trading days?

Twenty trading days correspond to approximately one calendar month. This window is:

- **Short enough** to respond quickly to changing conditions (e.g., a sudden spike in uncertainty following a shock).
- **Long enough** to avoid being dominated by single-day outliers that do not reflect a genuine change in the volatility regime.

Why multiply by $\sqrt{252}$?

This is the standard *annualisation factor* for daily volatility. There are approximately 252 trading days per year. Because variance (the square of volatility) scales linearly with time, and volatility is the square root of variance:

$$\sigma_{\text{annual}} = \sigma_{\text{daily}} \times \sqrt{252} \tag{3}$$

Intuition check

If SPY has a daily standard deviation of approximately 1%, then the annualised volatility is $1\% \times \sqrt{252} \approx 15.9\%$. This matches the commonly cited long-run SPY volatility of ∼15–16% per year. During market stress (2008, 2020), daily vol can reach 3–4%, implying annualised vol of 47–63%.

Listing 2: Realised volatility computation in features.py

```
df["vol_20d"] = df["ret"].rolling(20).std() * np.sqrt(252)
```

4.3 The 200-Day Moving Average

The 200-day *simple moving average* is the arithmetic mean of the 200 most recent closing prices:

$$\text{MA}_{200}(t) = \frac{1}{200} \sum_{i=0}^{199} P_{t-i} \tag{4}$$

Why 200 days?

Two hundred trading days correspond to approximately one full business year (52 weeks × ∼4.8 trading days). This is the horizon at which large institutional capital reallocates in response to structural changes: full earnings cycles, central bank policy shifts, and macroeconomic expansion or contraction.

The 200-day MA is the most widely-watched technical indicator in institutional finance, which creates a *self-reinforcing dynamic*: large capital flows respond to it, making it a meaningful equilibrium level that genuinely matters for price behaviour.

Why a simple moving average (SMA) rather than an exponential moving average (EMA)?

An EMA weights recent observations more heavily, making it more reactive. For a regime framework designed to capture *structural, long-term* trend states rather than short-term momentum, the SMA's uniform weighting is preferable. It is less susceptible to being flipped by short-lived price spikes that carry no structural meaning.

Listing 3: 200-day MA computation in features.py

```
df["ma_200"] = df["price"].rolling(200).mean()
```

4.4 Trend Distance

The *trend distance* is the normalised deviation of the current price from its long-term equilibrium:

$$\delta_t = \frac{P_t - \mathrm{MA}_{200}(t)}{\mathrm{MA}_{200}(t)} \tag{5}$$

This feature is not used for regime classification directly. The sign of the price-minus-MA difference handles that. Trend distance is used in the **confidence score** to measure how convincingly the trend signal is expressed. A price 15% above its MA is a much stronger trend signal than one 0.5% above it, even though both classify as `UP`.

Listing 4: Trend distance computation in features.py

```
df["trend_dist"] = (df["price"] - df["ma_200"]) / df["ma_200"]
```

5 Discretisation: Continuous to Categorical

Discretisation means converting a continuous numerical variable into a small number of categorical states. This is a deliberate design choice.

5.1 Why Discretise?

Continuous regime models (e.g., Hidden Markov Models, Gaussian Mixture Models) require parameter estimation via expectation-maximisation or Bayesian inference. These approaches can be unstable out-of-sample, are sensitive to initialisation, and produce regime *probabilities* that are harder to act on than a clear label.

Discrete, rule-based regimes are **robust, interpretable, and stable**. The cost is that fine gradations within a state are lost. For the purpose of structural situational awareness, that cost is acceptable, and the confidence score recovers some of the lost granularity.

5.2 Trend State

The trend state is derived from a simple price comparison:

$$\text{TrendState}(t) = \begin{cases} \texttt{UP} & \text{if } P_t > \text{MA}_{200}(t) \\ \texttt{DOWN} & \text{if } P_t < \text{MA}_{200}(t) \\ \texttt{NEUTRAL} & \text{otherwise} \end{cases} \tag{6}$$

In practice, price almost never sits exactly on the MA, so `NEUTRAL` trend is rarely triggered by this comparison. It exists to handle edge cases cleanly and prevent unclassified rows.

5.3 Volatility State: Calibration and Quantiles

What is a quantile?

The q-th *quantile* (or *percentile*) of a dataset is the value below which a fraction q of all observations fall. The 33rd percentile of volatility, for instance, is the level below which one-third of all observed volatility values lie.

Threshold estimation

The thresholds are estimated **exclusively on the calibration window (2000–2020)**, then stored as fixed scalar constants:

$$v_{\text{low}} = Q_{0.33}\left(\sigma_{20}\big|_{2000-2020}\right) = 10.43\% \tag{7}$$

$$v_{\text{high}} = Q_{0.66}\left(\sigma_{20}\big|_{2000-2020}\right) = 16.84\% \tag{8}$$

$$v_{\text{mid}} = \frac{v_{\text{low}} + v_{\text{high}}}{2} = 13.63\% \tag{9}$$

The volatility state for every day in the full dataset is then:

$$\text{VolState}(t) = \begin{cases} \texttt{LOW} & \text{if } \sigma_{20}(t) \leq v_{\text{low}} \\ \texttt{MEDIUM} & \text{if } v_{\text{low}} < \sigma_{20}(t) \leq v_{\text{high}} \\ \texttt{HIGH} & \text{if } \sigma_{20}(t) > v_{\text{high}} \end{cases} \tag{10}$$

Listing 5: Threshold calibration and vol state assignment in method1.py

```python
calib     = df.loc[calibration_start:calibration_end]
low_vol   = calib["vol_20d"].quantile(0.33)    # 10.43%
high_vol  = calib["vol_20d"].quantile(0.66)    # 16.84%
mid_vol   = (low_vol + high_vol) / 2.0         # 13.63% -- stored as constant

df["vol_state_raw"] = pd.cut(
    df["vol_20d"],
    bins=[-np.inf, low_vol, high_vol, np.inf],
    labels=["LOW", "MEDIUM", "HIGH"],
).astype(str)
```

5.4 Look-Ahead Bias: The Critical Design Constraint

What is Look-Ahead Bias?

Look-ahead bias (also called *future leak*) occurs when a model uses information that would not have been available at the time a decision was made. A model with look-ahead bias will appear to perform well historically but will **fail in live trading** because it "knew" things it could not have known.
This is the single most common and most damaging error in quantitative backtesting.

In this framework, look-ahead bias is avoided as follows:

1. The thresholds v_{low}, v_{high}, v_{mid} are computed on the in-sample period (2000–2020) only.
2. They are stored as fixed scalar constants.
3. They are applied to the entire dataset, including the out-of-sample period 2021–2026, **without any re-estimation**.

The out-of-sample data *never* influences the threshold values. The confidence score formula uses only these calibration-period constants, not any statistics computed on the full dataset.

Concrete numerical example from the actual data.

Imagine you are sitting at your desk on **1 March 2005**. You want to classify that day's regime. SPY is at $121.23, above its 200-day MA, so the trend is `UP`. The 20-day realised volatility is **10.61%**.

The only question is: does 10.61% fall in the `LOW` bucket or the `MEDIUM` bucket? That depends on where the threshold between them is set.

Scenario A: Correct (thresholds computed on 2000–2020 only)

Using only data up to and including 2020, the 33rd percentile of volatility is **10.43%**.

$$10.61\% > 10.43\% \;\Rightarrow\; \text{VolState} = \texttt{MEDIUM} \;\Rightarrow\; \text{Regime} = \texttt{RISK_ON_FRAGILE}$$

Scenario B: Wrong (thresholds computed on the full 2000–2026 dataset)

Now suppose you made a mistake and computed your threshold using all available data, including everything from 2020 onwards. The COVID crash in March 2020 generated daily volatility above 60% annualised. The years 2021–2026 were also unusually volatile compared to the pre-2020 average. Adding all those extreme values into the distribution shifts things: the 33rd percentile rises slightly from 10.43% to **10.65%**.

$$10.61\% \leq 10.65\% \;\Rightarrow\; \text{VolState} = \texttt{LOW} \;\Rightarrow\; \text{Regime} = \texttt{RISK_ON}$$

The same day. The same price. The same volatility reading. Two different regime labels, purely because the threshold was computed differently.

Why Scenario B is cheating: On 1 March 2005, COVID had not happened yet. It would not happen for another 15 years. There is absolutely no way you could have known on that date that 2020 would be one of the most volatile years in market history. The threshold of 10.65% carries information from events that had not yet occurred. Using it to classify a 2005 trading

day means the model is reaching forward in time, borrowing knowledge from the future to relabel the past. That is look-ahead bias.

Why it matters: `RISK_ON` has a Sharpe ratio of 0.35 and `RISK_ON_FRAGILE` has a Sharpe of 0.88. By incorrectly classifying this day (and 9 other days in 2005 alone) as `RISK_ON` instead of `RISK_ON_FRAGILE`, the backtested Sharpe for `RISK_ON` gets inflated. The model looks better on paper than it actually is. In live trading, where you only know what has happened so far, the 10.65% threshold does not exist yet, and the performance you saw in the backtest will not repeat.

Fixing the threshold at 10.43% (in-sample only) and treating it as a constant from that point on closes this problem entirely.

5.5 Volatility Smoothing: Rolling Majority Vote

Even after discretisation, the raw volatility state can *flicker* when realised vol sits near a boundary between states. On consecutive days, the state might alternate between `MEDIUM` and `HIGH` purely because vol is oscillating by a fraction of a percent around the threshold. These transitions carry no structural meaning.

To address this, a **5-day rolling majority vote** is applied:

$$\text{VolState}_{\text{smooth}}(t) = \text{mode}(\text{VolState}_{\text{raw}}(t-4), \ldots, \text{VolState}_{\text{raw}}(t)) \tag{11}$$

The smoothed state on day t is the most frequent raw state over the past 5 days. A state must hold as the majority over a full week before it takes effect.

Listing 6: Majority vote smoothing in method1.py

```
def smooth_state(series, window=5):
    mapping          = {"LOW": 0, "MEDIUM": 1, "HIGH": 2}
    inverse_mapping  = {v: k for k, v in mapping.items()}
    numeric          = series.map(mapping)
    smoothed         = numeric.rolling(window, min_periods=1).apply(
                           lambda x: pd.Series(x).value_counts().idxmax(),
                           raw=False)
    return smoothed.map(inverse_mapping)
```

6 Regime Classification

6.1 The Classification Grid

With two trend states (`UP`, `DOWN`) plus `NEUTRAL`, and three volatility states, there are theoretically nine combinations. The classification grid maps them to six named regimes:

Table 2: The complete regime classification grid

	Vol: LOW	Vol: MEDIUM	Vol: HIGH
Trend: UP	`RISK_ON`	`RISK_ON_FRAGILE`	`BULL_EXPANSION`
Trend: DOWN	`RISK_OFF_TRANS.`	`RISK_OFF_TRANS.`	`RISK_OFF`
Trend: NEUTRAL	`NEUTRAL`	`NEUTRAL`	`NEUTRAL`

Listing 7: The classify function in method1.py: complete logic

```
def _classify_row(trend: str, vol: str) -> str:
    if trend == "UP"   and vol == "LOW":    return "RISK_ON"
    if trend == "UP"   and vol == "MEDIUM": return "RISK_ON_FRAGILE"
    if trend == "UP"   and vol == "HIGH":   return "BULL_EXPANSION"
    if trend == "DOWN" and vol == "HIGH":   return "RISK_OFF"
    if trend == "DOWN":                     return "RISK_OFF_TRANSITION"
    return "NEUTRAL"  # trend == "NEUTRAL", any vol
```

6.2 The Six Regimes Defined

RISK_ON — UP trend + LOW volatility

The ideal risk-taking environment. The long-term trend is positive and volatility is low, meaning capital flows are orderly and investor conviction is high. Drawdowns tend to be shallow and short-lived. Risk premia compress.

Characteristic period: 2017: the entire year classified as RISK_ON. SPY advanced steadily with minimal day-to-day movement. 251 trading days of uninterrupted RISK_ON.

Backtest (2000–2026): Annualised return 3.9%, volatility 11.2%, Sharpe 0.35, max drawdown −22.3%, 1,962 days (30.7%). *See Section 9 for the full explanation of why the raw return appears low.*

RISK_ON_FRAGILE — UP trend + MEDIUM volatility

The trend remains positive but stress is building. Volatility has risen into the middle tercile, indicating investors are becoming less certain and position-taking is less orderly. Returns can still be positive but the downside risk is becoming asymmetric.

Characteristic period: Late 2021: markets were rallying to all-time highs with rising inflation anxiety. 137 of 252 trading days in 2021.

Backtest: Annualised return 11.7%, volatility 13.2%, Sharpe 0.88, max drawdown −22.0%, 1,989 days (31.2%).

BULL_EXPANSION — UP trend + HIGH volatility

An uptrend with high volatility captures two scenarios: violent post-crisis recoveries (the 2020 COVID rebound, where price rose explosively with enormous daily swings), and late-cycle momentum-driven rallies. Returns are high but so is the drawdown risk. If the trend reverses, the exit will be violent.

Characteristic period: April to August 2020: 95 days of BULL_EXPANSION as SPY recovered from the COVID crash at extreme speed with daily moves of 2–4%.

Backtest: Annualised return 24.1%, volatility 16.5%, Sharpe 1.46 (highest of all regimes), max drawdown −12.7%, 598 days (9.4%).

NEUTRAL — Price near 200-day MA

Price is oscillating near the 200-day MA with no dominant structural direction. Capital flows are balanced; the market is reassessing. Characterised by choppy price action and low signal quality.

Backtest: Effectively zero days observed ($< 0.1\%$) in the full SPY history. Price spends almost no time exactly at the MA — it either breaks above and trends, or breaks below.

RISK_OFF_TRANSITION **DOWN trend + LOW or MEDIUM volatility**

The most dangerous and underappreciated regime. The trend has turned negative (the market is below its 200-day MA) but volatility has not yet spiked. Investors often mistake this for a buying opportunity because everything *looks* calm. In reality, capital is quietly de-risking. This is frequently the regime immediately before a full RISK_OFF event.

Characteristic period: The 36 days preceding the worst of the 2008 GFC collapse. Markets were below the MA but volatility had not yet exploded. Those who recognised this signal avoided the worst of what followed.

Backtest: Annualised return −10.2%, volatility 19.5%, Sharpe −0.52, max drawdown −30.3%, 340 days (5.3%).

RISK_OFF **DOWN trend + HIGH volatility**

The full risk-off environment. Forced selling and deleveraging are underway. Correlations between assets spike, liquidity in secondary markets dries up, and bid-ask spreads widen. Investors still positioned for risk are being forced out. Capital preservation is the only objective.

Characteristic period: 2008: 217 of 252 trading days classified as RISK_OFF. The remaining 36 were RISK_OFF_TRANSITION.

Backtest: Annualised return −1.1%, volatility 31.5%, Sharpe −0.04, max drawdown −60.0%, 1,494 days (23.4%).

7 Persistence Filter

7.1 The Problem: Threshold-Boundary Noise

If a regime were assigned on every day based purely on that day's trend and volatility states, the resulting sequence would be very noisy near state boundaries. When realised volatility oscillates around the 10.43% or 16.84% threshold, the raw vol state can flip between adjacent levels on consecutive days. This produces one- or two-day "regime changes" that carry no structural meaning.

7.2 The 3-Day Confirmation Rule

A candidate new regime must appear on at least **3 consecutive trading days** before it is accepted:

$$\text{confirmed}[t] = \begin{cases} \text{raw}[t] & \text{if raw}[t] \text{ has held for } \geq 3 \text{ consecutive days} \\ \text{confirmed}[t-1] & \text{otherwise} \end{cases} \tag{12}$$

Listing 8: Persistence filter implementation in method1.py: NumPy loop to avoid SettingWithCopy-Warning

```
def confirm_regime(series: pd.Series, min_days: int = 3) -> pd.Series:
    values    = series.to_numpy(dtype=object)
    confirmed = values.copy()
    count     = 1

    for i in range(1, len(values)):
        if values[i] == values[i - 1]:
            count += 1
```

```
        else:
            count = 1
        if count < min_days:
            confirmed[i] = confirmed[i - 1]

    return pd.Series(confirmed, index=series.index, name=series.name)
```

7.3 Empirical Result

Verified on the full 2000–2026 dataset:

- Total regime blocks: **281**
- Minimum block length: **3 days** (zero blocks shorter than 3 days)
- Maximum block length: **296 days** (RISK_ON, 2003–2004)
- Mean block length: **22.7 days**

7.4 No Look-Ahead in the Filter

The confirmation rule is strictly *causal*, meaning it only looks backward. On day t, the confirmed regime is determined using only data from days 0 through t. No future information is used. The cost is a maximum 2-day delay in recognising a genuine regime change. This is an intentional and acceptable trade-off for the noise reduction it provides.

Implementation note

The filter is implemented by converting the pandas Series to a NumPy array before the loop. This avoids the `SettingWithCopyWarning` that arises from `.iloc[i] =` writes inside a for-loop on a pandas Series, and is significantly faster for large arrays because it bypasses pandas' indexing overhead.

8 Confidence Score

8.1 What Confidence Measures

The confidence score is a number in $[0, 1]$ measuring how clearly and convincingly the market is expressing the detected regime. It answers: *"Given that we are in this regime, how strongly is the market expressing it?"*

It is **not** a probability of correctness. The regime label is always the one produced by the classification logic. Confidence is a measure of signal *conviction*.

High vs. Low Confidence

High confidence (e.g. 0.90): Price is far above the MA, vol is well inside its bucket (far from any boundary), and the regime has been stable for weeks.

Low confidence (e.g. 0.30): Price is barely above the MA, vol is right at a threshold, and the regime was just confirmed yesterday.

8.2 Component 1: Trend Score (Weight: 40%)

The trend score measures how far the current price is from its 200-day MA, normalised at a 5% conviction threshold:

$$S_{\text{trend}}(t) = \text{clip}\left(\frac{|\delta_t|}{0.05},\ 0,\ 1\right) \tag{13}$$

where δ_t is the trend distance defined in Equation (5) and $\text{clip}(x, 0, 1) = \max(0, \min(1, x))$.

A price 5% or more from its MA scores 1.0. A price exactly at the MA scores 0. The 5% level is the point at which institutional investors would consider the trend direction structurally confirmed.

Trend score receives the highest weight (40%) because trend direction is the primary regime driver. Whether you are above or below the MA is the most consequential single fact about the current regime.

Listing 9: Trend score in method1.py

```
trend_score = np.clip(np.abs(df["trend_dist"]) / 0.05, 0, 1)
```

8.3 Component 2: Volatility Score (Weight: 30%)

The volatility score measures how far the current vol is from the midpoint between the two calibration thresholds. In other words, it measures how unambiguously the vol falls inside its assigned state:

$$\Delta v = v_{\text{high}} - v_{\text{low}} = 6.41\% \tag{14}$$

$$S_{\text{vol}}(t) = \text{clip}\left(1 - \frac{|\sigma_{20}(t) - v_{\text{mid}}|}{\Delta v},\ 0,\ 1\right) \tag{15}$$

Volatility exactly at $v_{\text{mid}} = 13.63\%$ (perfectly centred) scores 1.0. Volatility right at $v_{\text{low}} = 10.43\%$ or $v_{\text{high}} = 16.84\%$ (ambiguous boundary) scores 0.

Critical: All values in this formula (v_{mid}, Δv) are constants from the calibration window. Out-of-sample volatility values do not affect them.

Listing 10: Volatility score in method1.py: using calibration constants only

```
vol_range = high_vol - low_vol          # 0.0641 -- calibration constant
vol_dist  = np.abs(df["vol_20d"] - mid_vol) / vol_range
vol_score = np.clip(1.0 - vol_dist, 0, 1)
```

8.4 Component 3: Duration Score (Weight: 30%)

The duration score rewards regimes that have been stable for longer:

$$S_{\text{dur}}(t) = \text{clip}\left(\frac{d(t)}{20},\ 0,\ 1\right) \tag{16}$$

where $d(t)$ is the number of consecutive days the current confirmed regime has been active. A regime that has lasted ≥ 20 days scores 1.0. A regime on its first day scores 0.05.

Rationale: A regime that has persisted for three weeks is more reliable than one confirmed yesterday. Duration is evidence of structural stability, not noise.

Listing 11: Duration score in method1.py

```
blocks          = (df["regime"] != df["regime"].shift()).cumsum()
duration        = df.groupby(blocks).cumcount() + 1
duration_score  = np.clip(duration / 20, 0, 1)
```

8.5 Full Formula

The three components are combined as a weighted sum:

$$\boxed{C(t) = 0.4 \cdot S_{\text{trend}}(t) + 0.3 \cdot S_{\text{vol}}(t) + 0.3 \cdot S_{\text{dur}}(t)} \tag{17}$$

Table 3: Confidence score component weights and rationale

Component	Weight	Rationale
S_{trend}	40%	Trend is the primary regime driver; its conviction is the most economically meaningful input
S_{vol}	30%	Proximity to a boundary is the main source of classification uncertainty
S_{dur}	30%	Regime persistence is structural evidence; recently-confirmed regimes deserve less trust

Empirical distribution (SPY 2000–2026): mean 0.631, median 0.684, range $[0.027, 1.000]$. Confidence drops sharply at every regime transition and rebuilds over the following weeks as the new regime stabilises.

9 Performance Statistics

9.1 How Statistics Are Computed

For each regime, performance statistics are computed on the subset of trading days where that regime was active. This produces a collection of **non-consecutive daily observations** spread across the entire 2000–2026 period.

9.2 Annualised Return

$$\bar{r}_{\text{ann}} = \bar{r}_{\text{daily}} \times 252 = \frac{1}{N} \sum_{t \in \text{regime}} r_t \times 252 \tag{18}$$

This is the expected annualised return if every trading day had the regime's average daily return. It does not account for compounding across the fragmented, non-consecutive days.

9.3 Annualised Volatility

$$\sigma_{\text{ann}} = \sigma_{\text{daily}} \times \sqrt{252} = \text{std}(\{r_t : t \in \text{regime}\}) \times \sqrt{252} \tag{19}$$

9.4 Sharpe Ratio

$$\text{Sharpe} = \frac{\bar{r}_{\text{ann}}}{\sigma_{\text{ann}}} \tag{20}$$

No risk-free rate is deducted (gross Sharpe). A Sharpe above 1.0 is considered good; above 1.5 is excellent; negative means the regime loses money on average.

9.5 Maximum Drawdown

$$\text{CumRet}(t) = \prod_{i \le t,\, i \in \text{regime}} (1 + r_i) \tag{21}$$

$$\text{Peak}(t) = \max_{i \le t} \text{CumRet}(i) \tag{22}$$

$$\text{DD}(t) = \frac{\text{CumRet}(t) - \text{Peak}(t)}{\text{Peak}(t)} \tag{23}$$

$$\text{MaxDD} = \min_{t} \text{DD}(t) \tag{24}$$

Maximum drawdown measures the largest peak-to-trough decline within the regime observation set.

9.6 The RISK_ON Return Paradox: Full Explanation

The aggregate statistics show RISK_ON with an annualised return of only 3.9% and a Sharpe of 0.35, while RISK_ON_FRAGILE shows 11.7% and 0.88. At first glance this appears wrong. The explanation has two parts.

Part 1: Economic Reality

Low-volatility uptrends include extended periods of mature bull markets where price moves sideways within a tight range. In 2017, SPY advanced the entire year with almost no daily movement, and the mean daily return was barely above zero. The 2004–2006 period was similar. These are genuinely low-return, low-risk environments. The RISK_ON label is *correct*: they are risk-on environments. But "risk-on" means risk-taking is safe and appropriate, not that you will earn high returns.

Part 2: Statistical Artefact

The fragmented, non-consecutive grouping suppresses compounding. When RISK_ON returns are broken into coherent calendar sub-periods, the picture changes completely:

Table 4: RISK_ON performance by calendar sub-period (SPY)

Sub-period	Ann. Return	Ann. Vol	Sharpe
2000–2007	+18.7%	14.1%	1.32
2008–2014	+18.1%	28.2%	0.64
2015–2020	+21.7%	15.4%	1.41
2021–2026	+48.9%	22.5%	2.18

Within every coherent sub-period, RISK_ON produces Sharpe ratios of 0.64 to 2.18, which is consistently the best risk-adjusted performance of any regime. The 3.9% aggregate figure is an

artefact of averaging mean daily returns across 1,962 fragmented days spanning 26 years of varying base levels.

Correct Interpretation of RISK_ON

The right metrics for RISK_ON are **Sharpe ratio** and **maximum drawdown**, not raw annualised return. RISK_ON is precisely what its label says: the structurally safest, most capital-efficient environment. It earns modest returns *with minimal risk*.

10 Empirical Results

10.1 Full Backtest Statistics (SPY, Oct 2000 – Mar 2026)

Table 5: Per-regime performance statistics

Regime	Ann. Ret.	Ann. Vol	Sharpe	Max DD	Days	Share
RISK_ON	+3.9%	11.2%	0.35	−22.3%	1,962	30.7%
RISK_ON_FRAGILE	+11.7%	13.2%	0.88	−22.0%	1,989	31.2%
BULL_EXPANSION	+24.1%	16.5%	1.46	−12.7%	598	9.4%
NEUTRAL	—	—	—	—	—	—
RISK_OFF_TRANS.	−10.2%	19.5%	−0.52	−30.3%	340	5.3%
RISK_OFF	−1.1%	31.5%	−0.04	−60.0%	1,494	23.4%

10.2 Crisis Year Validation

The most important test for any regime model is whether it correctly classifies periods that history has already judged. No parameters were tuned to these outcomes.

Table 6: Model output during major historical events

Year	Event	Model output	Verdict
2008	Global Financial Crisis	217d RISK_OFF, 36d RISK_OFF_TRANS.	Correct
2017	Quietest bull market	251d RISK_ON (entire year)	Correct
2020	COVID crash + recovery	57d RISK_OFF → BULL_EXP → FRAGILE	Correct
2022	Fed rate-hike bear	199d RISK_OFF, 29d BULL_EXP	Correct

10.3 Regime Duration Statistics

Table 7: Regime block duration statistics (281 total blocks)

Regime	Blocks	Mean (days)	Median	Max
RISK_ON	73	26.9	19	296
RISK_ON_FRAGILE	105	18.9	19	81
BULL_EXPANSION	42	14.2	12	42
RISK_OFF_TRANS.	31	11.0	7	38
RISK_OFF	30	49.8	32	245

RISK_OFF has the longest mean duration (49.8 days). Once full risk-off is confirmed, it tends to persist. RISK_OFF_TRANSITION is the shortest at 11 days. It either escalates to RISK_OFF or resolves upward quickly.

10.4 Transition Probability Matrix

On any given day, the empirical probability of moving from one regime to another:

Table 8: Empirical daily transition probabilities (rows sum to 100%)

From \ To	R_ON	R_ON_F	BULL	R_OFF_T	R_OFF
RISK_ON	**96.3%**	3.2%	0.2%	0.4%	0.0%
RISK_ON_FRAGILE	3.6%	**94.8%**	1.0%	0.5%	0.2%
BULL_EXPANSION	0.2%	5.0%	**93.0%**	0.2%	1.7%
RISK_OFF_TRANS.	0.3%	3.2%	0.9%	**90.9%**	4.7%
RISK_OFF	0.0%	0.1%	1.1%	0.9%	**98.0%**

Two observations stand out. First, **RISK_ON never transitions directly to RISK_OFF (0.0%)**. The market always deteriorates through intermediate regimes first, which gives you time to react. Second, the RISK_OFF_TRANSITION to RISK_OFF transition runs at 4.7% per day. That sounds small, but over 20 days of RISK_OFF_TRANSITION the cumulative probability of entering full RISK_OFF becomes substantial.

11 Limitations

No model is complete without an honest accounting of what it cannot do.

Lagging by construction. The 200-day MA and 20-day volatility window are backward-looking. The 3-day persistence filter adds a further minimum 2-day delay. Regime changes are confirmed after the fact, not predicted. A market crash that begins and ends within three days would never be captured.

Calibration window dependency. Thresholds are fixed to SPY's volatility distribution from 2000–2020. If the structural volatility of the market shifts materially (e.g., a sustained period of ultra-low volatility following central bank intervention), the thresholds may no longer correctly partition the distribution. Periodic recalibration is advisable for live use.

Single-asset calibration. Designed for a broad equity index ETF. Applying to individual stocks,

fixed income, or commodities requires recalibrating the volatility thresholds to that asset's distribution.

Gross Sharpe ratios. No risk-free rate is deducted. Subtract the annualised short rate (e.g., 3-month T-bill) to compute excess-return Sharpe ratios. This will reduce all figures by approximately 0.2–0.5 depending on the interest rate environment.

Fragmented regime statistics. Computing performance statistics over non-consecutive regime days produces artefacts, particularly in annualised return and max drawdown. See Section 9.6 for the full analysis. These metrics should be interpreted with the sub-period breakdown in mind.

No transaction costs. Backtest statistics assume frictionless execution. Any strategy that rebalances on regime transitions will incur transaction costs, slippage, and market impact that have not been modelled.

12 How to Use This Framework

12.1 Running the Model

```
# Default: SPY
python run.py

# Any Yahoo Finance ticker with sufficient history
python run.py QQQ
```

The pipeline downloads data, computes features, applies the regime detection, prints the live signal, generates all visualisations, and exports both CSV files.

12.2 Interpreting the Live Signal

Table 9: Practical interpretation guide for portfolio managers

Regime	Confidence	Interpretation
RISK_ON	$> 70\%$	Full risk exposure appropriate
RISK_ON	$< 50\%$	Newly confirmed. Size conservatively until stability builds
RISK_ON_FRAGILE	Any	Maintain exposure with tighter stop levels
BULL_EXPANSION	Any	High return potential; size for high vol; have an exit plan
NEUTRAL	Any	Reduce directional exposure; wait for clarity
RISK_OFF_TRANS.	Any	Begin reducing risk; do not wait for full RISK_OFF confirmation
RISK_OFF	$> 70\%$	Defensive positioning; capital preservation is the only objective
RISK_OFF	$< 50\%$	Possible reversal in progress; monitor closely

12.3 Project Structure

File	Purpose
`data.py`	Downloads and validates price data via `yfinance`
`features.py`	Computes log returns, 20d vol, 200d MA, trend distance
`method1.py`	Core engine: discretisation, classification, persistence filter, confidence score
`analytics.py`	Per-regime performance statistics and bar charts
`plotting.py`	All six visualisation types
`run.py`	End-to-end pipeline entry point; accepts ticker as CLI argument